Jiale (Caleb) Liu

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EDUCATION & COURSEWORK

Rensselaer Polytechnic Institute, School of Computer Science and Business – Troy, NY **May 2028**
Bachelor of Science in Computer Science and Business Analytics Minor: Finance Cumulative GPA: 3.64/4.00

Computer Science:, Computer Architecture and Operating Systems, Rensselaer Center for Open Source
Math: Foundations of Computer Science, Intro to Linear Algebra

TECHNICAL SKILLS

Next, Python, MongoDB –Linux, Node, React, Vue, Java, Javascript, PyTorch, C, C#, C++, SQL, Tailwind CSS, Git, GitLab CI, AWS

PROFESSIONAL EXPERIENCE

RPI AI/ML Biotechnology Robotics Research | AI/ML and Data Pipeline Researcher | Troy, NY **September 2025 – Present**
- Architect PyTorch Graph Neural Network (GNN) with 3 feature extraction pipelines (Intrinsic Mode, Hilbert-Huang, Hjorth) to boost predictive accuracy by 84%, and standardize brain wave data gathering protocols
- Engineer asynchronous control interface that decoupled neural decoding from motor threads, executing real-time 3D robotic hand trajectories with sub-50ms latency.

HackRPI | Director of Technology | Troy, NY **September 2025 – Present**
- Lead a 8-developer team across 3 Agile/Scrum sprints to build and open-source Next.js website and UI, coordinating domain migration and cross-club partnerships to boost platform traffic by 43%
- Re-architect frontend responsiveness across mobile viewports using Next.js, TypeScript, Tailwind CSS, and React, reducing mobile bounce rate by 75% and ensuring seamless cross-device performance
- Engineer accessible UI layouts using HTML5, eliminating layout overlap bugs to achieve WCAG 2.1 accessibility compliance
- Develop an interactive 3D web game using Three.js, Socket.io, Blender, building custom rig animation tools and dynamic lighting pipelines to boost user session duration by 30%

RPI Union | Production Fullstack Engineer | Troy, NY **September 2025 – Present**
- Engineer and deploy 15 end-to-end React/Node.js/Vite.js features across the full stack, resolving high-priority system bugs to improve application stability and optimize API latency for 10,000+ users

MoreNBest Decor | Automation Lead | Columbia, SC **March 2023 – September 2025**
- Replaced manual seller-distributor communications with autonomous updates by building autonomous pipelines for inventory reports, sales analysis, and email ingestion- expanding operational automation coverage by 90%
- Engineered a temporal dynamic pricing model integrated with Amazon Seller Central, leveraging real-time inventory telemetry to optimize product margins and increase profit margins by 1-3%
- Developed an automated data extraction engine using Gmail API, Selenium, SMTP, and IMAP to parse sender metadata, body text, and attachments, processing 10,000+ purchase records daily for actionable insights

PROJECTS

Blockchain Project | Personal | Troy, NY **July 2026**
- Developed a high-throughput blockchain application using Python and Flask, constructing RESTful API endpoints and rigorous unit testing pipelines that ensured 100% consensus accuracy during stress testing

Plantilir | HackRPI | Troy, NY **November 2025**
- Engineered a touchless spatial interaction engine by mapping 2D video telemetry to 3D space via MediaPipe, tracking 500+ hand and facial landmark points to control dynamic visual models in Three.js

Neural Language Model | Personal | Troy, NY **September 2025 – October 2025**
- Engineered an end-to-end NLP pipeline using NLTK and PyTorch to vectorize complex literature datasets, building a multi-layer Feed-Forward Neural Network that predicted next-token sequences with 61% accuracy

NapCast| Cornell BigRedHacks | Ithaca, NY **September 2025**
- Engineered an automated AI podcast generator trained on scraped Spotify audio datasets, deploying a multi-model PyTorch architecture to generate contextual scripts and deliver customizable multi-voice narration, with 1:1 real-time generation latency

Autonomous Video Content Generator | Personal | Troy, NY **June 2025 – August 2025**
- Designed an automated end-to-end media generation pipeline in Docker, leveraging PRAW (Reddit API), OpenAI Whisper, and FFmpeg to scrape trending stories and auto-render narrated YouTube Shorts at a scale of 140 videos/day